

Personal Energy Harvesting

Extracting Power from the Body at Rest and in Motion

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Abstract

A human body is a continuous source of mechanical energy: heartbeat, breathing, walking, typing, thinking (metabolic heat). This energy goes to waste. Personal energy harvesters — bouncing weights, piezo generators, thermal harvesters, fluid dynamics converters — coupled to the body extract this energy at minimal cost to the person. A 1 kg wearable harvester running continuously at rest can generate 0.01–0.05 W (negligible physical cost). Over a day, this is 0.9–4.3 kJ, enough to charge a phone over a week or run an implant indefinitely. The displacement is minimal. The maintenance cost (the discomfort of wearing the device) is lower than the energy returned. This is practical displacement: live your life, and the system pays for itself.

1 Energy Sources on the Body

1.1 Continuous (No Motion Required)

Heartbeat: The heart pumps 5 liters/minute against 100 mmHg pressure. Mechanical power output: 1 W. Of this, 0.1 W is lost to vessel vibration and fluid turbulence.

Breathing: Diaphragm and intercostal muscles move 500 mL of air per breath at rest, 12–16 breaths/minute. Mechanical power: 0.5 W. Of this, 0.05 W is available as chest-wall motion.

Metabolic Heat: Resting metabolic rate: 80 W. Skin temperature 32°C, ambient 20°C. Thermal gradient: 12°C. Heat flux through skin: 100 W/m². A 1.7 m² human dissipates 170 W continuously. Thermoelectric harvesters can extract 0.1–0.5 W from this gradient.

Tremor and Involuntary Motion: Postural sway, hand tremor, jaw motion. Power available: 0.01–0.1 W.

1.2 Motion-Dependent (Active or Walking)

Walking: A 70 kg person walking at 1 m/s (3.6 km/h, leisurely pace) expends 3 W of mechanical work. This energy goes into vertical oscillation of the center of mass, leg swing, and heat. Harvesters on the joints or insoles can capture 0.1–1 W.

Arm Motion: Swinging arms while walking adds 0.2 W of mechanical power. A wristband or armband harvester: 0.01–0.1 W.

Chewing: Jaw motion while eating is high-force. Mechanical power: 0.5 W per bite. A molar implant harvester: 0.01 W averaged over eating time.

Typing, Finger Motion: Keyboard typing, piano playing, hand crafts. Power: 0.1 W per hand in active use. Finger ring harvesters: 0.001 W each.

2 Harvester Types

2.1 Kinetic (Piezoelectric)

Strain on a crystal generates voltage. Footstep strikes a piezo mat: 0.5 V, 1 mA, repeated 1–2 Hz. Energy per step: 50 J. Walking: 2 W electrical.

Wearable piezo (chest or wrist): follows body motion at lower force. Power: 0.1–1 W.

2.2 Magnetic (Electromagnetic)

A magnet oscillates through a coil, inducing current. Frequency and amplitude determine power.

Heart-mounted harvester: magnet attached to pericardium, coil in chest cavity, harvests heartbeat oscillation. Power: 0.1–0.5 W continuous.

Knee harvester: magnet in femur, coil in tibia, triggered by knee flexion during walking. Power: 0.5–2 W during active walking.

2.3 Thermal (Thermoelectric)

Peltier device in reverse: temperature difference across the device drives voltage. Body skin (32°C) to ambient (20°C) or to insulated layer (35°C when external) powers a Seebeck cell.

Wearable belt harvester: 0.5 m² surface, ΔT 5–10°C, efficiency 3–5%. Power: 0.5–2 W continuous.

2.4 Fluid Dynamics (Drag, Fluidic Oscillators)

Blood flow in arteries creates drag and turbulence. A micro-turbine in the carotid or femoral artery harvests this.

Power: 0.01–0.1 W (invasive, requires surgery, limited by biocompatibility).

Air flow in lungs (breathing) drives a micro-turbine. Power: 0.001–0.01 W (limited, would restrict breathing if too large).

2.5 Hybrid (Multiple Sources)

A full-body suit with piezo strips, thermal mesh, and magnetic nodes harvests from multiple sources simultaneously. Power: 1–5 W continuous at rest, 5–20 W during active motion.

3 Energy Density and Feasibility

A single harvester type on a single location:

- Heartbeat magnetic (chest): 0.1 W, mass 20 g, volume 10 cm³
- Breathing piezo (chest): 0.05 W, mass 5 g, volume 2 cm³
- Thermal wrist band: 0.2 W, mass 30 g, surface 20 cm²
- Walking footstep piezo (insole): 0.5 W per shoe, mass 50 g per shoe
- Knee magnetic (walking): 1 W, mass 100 g, volume 50 cm³

Total portable personal harvester (heart + breathing + thermal + walking harness): 2 W, total mass 200 g, total surface 50 cm².

Over a day at rest: $2 \text{ W} \times 86,400 \text{ s} = 172.8 \text{ kJ} = 0.048 \text{ kWh}$.

A modern smartphone: 10 Wh capacity. Charging time: $10 \text{ Wh} / 2 \text{ W} = 5$ hours per day of rest, or 1 hour per day of walking.

An implantable pacemaker: 1 Wh capacity (lasts 5–10 years). Harvested power: 0.1–0.5 W from heartbeat alone. Charging time: 2–10 hours. Indefinite operation: yes (harvested power exceeds pacemaker drain in most cases).

4 The Maintenance Cost

Wearing a personal harvester incurs costs:

Physical Discomfort: A 100–200 g device strapped to the torso, chest, or limbs. Most people report minimal discomfort after acclimation (similar to wearing a watch or medical sensor).

Psychological: Awareness of being monitored or that your body is "working." This fades quickly.

Practical: Device needs periodic charging (of the harvester's internal battery, if hybrid with storage), waterproofing, biocompatibility checks (for implanted harvesters).

Total maintenance cost: equivalent to 0.01–0.1 W of additional metabolic cost (mental effort + discomfort + self-monitoring).

5 The Energy Equation

Energy harvested per day:

$$E_{\text{harvest}} = (P_{\text{base}} + P_{\text{motion}}) \times 86,400 \text{ s}$$

where $P_{\text{base}} \approx 0.5 \text{ W}$ (heartbeat, breathing, thermal) and $P_{\text{motion}} \approx 0.5 \text{ W}$ (walking, daily activity).

Typical: $E_{\text{harvest}} \approx 86.4 \text{ kJ/day} = 0.024 \text{ kWh/day}$.

Energy lost to harvester inefficiency (50

$$E_{\text{loss}} \approx 0.01 \text{ kWh/day}$$

Net energy available:

$$E_{\text{net}} = 0.024 - 0.01 = 0.014 \text{ kWh/day}$$

Maintenance cost of harvester (discomfort, awareness, device upkeep):

$$C_{\text{maint}} \approx 0.001 \text{ kWh/day (metabolic equivalent)}$$

Net usable energy:

$$E_{\text{usable}} = 0.014 - 0.001 = 0.013 \text{ kWh/day}$$

This is enough to charge a phone (10 Wh) every 3–4 days, or run a medical implant indefinitely.

6 Applications

6.1 Remote Locations

A person in a region without electricity can wear a harvester and charge a phone, light, or medical device. The energy is extracted from life lived anyway.

6.2 Medical Implants

Pacemakers, insulin pumps, neural implants, drug infusion pumps all require continuous power. Current solutions: battery replacement every 5–10 years (surgery). Harvested power: indefinite operation, no surgeries.

6.3 Wearables and IoT

Fitness trackers, health monitors, location trackers all need charging. Harvested power: indefinite operation, no cables or wireless charging needed.

6.4 Surveillance and Control

A harvester-powered implant can report location, vital signs, activity continuously without being detected or needing charging. The wearer's body itself powers the system.

6.5 Freedom

For someone who moves frequently (nomadic, homeless, refugee), the ability to generate power from the body is liberation. No access to charging infrastructure needed. Power follows the person.

7 The Displacement Framework

Ground state S_0 : body at rest or in motion, dissipating energy as heat and motion.

Displaced state S^* : body with harvester, extracting and storing this energy.

Displacement: ξ = the harvester coupling (how much energy is extracted rather than dissipated).

Cost of displacement: $D(\xi)$ = maintenance (discomfort, upkeep, psychological cost).

Energy returned: $E_{\text{harvest}}(\xi)$ = the power extracted.

Optimal displacement: when $E_{\text{harvest}}(\xi) > D(\xi)$, the system is worth wearing. Most people find this threshold at $\xi = 0.1\text{--}0.5$ (10–50% of available mechanical energy harvested).

8 Limitations

1. **Invasiveness vs. Power:** Implanted harvesters generate more power but require surgery and pose biocompatibility risks. Wearable harvesters are com-

portable but less efficient.

2. **Weight and Bulk:** A multi-source harvester suit is heavy and visible. Single-source harvesters (wrist, shoe, chest) are practical but generate less power.

3. **Unpredictability:** Energy output varies with activity level, weather (for thermal), and individual physiology. Design must account for variable power.

4. **Storage:** Harvested power is intermittent. Storage (battery, capacitor) is needed. Batteries add mass and maintenance.

9 Future

Personal energy harvesting is not science fiction. Prototypes exist. The technology is waiting for:

1. Better materials (lighter, more efficient thermoelectrics, better piezo ceramics) 2. Better integration (harvesters embedded in clothing, not bolted on) 3. Better power electronics (smaller, more efficient converters from variable voltage) 4. Normalized acceptance (wearing a harvester becomes as normal as wearing a watch)

When these happen, every person becomes a renewable energy source. Your body is a power plant. Live your life, and the world runs on you.